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(1) Research Program/Project/Study Title: DEVELOPMENT OF PAINT MIXER DEVICE WITH PAINTING APPARATUS	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation							
Municipality/City:	Roxas City						
Province:	Capiz						
Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:						
☐ Basic	☐ Pilot Testing						
☒ Applied	☐ Tech. Promotion/Commercialization						
(7) Sector/Commodity Construction and Hardware Supply Furniture Business DTI (Department of Trade and Industry) DepEd- TLE	(8) Discipline/Field of Specialization Drafting Architecture Civil Engineering						

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The researchers develop paint mixer device with painting apparatus for research study with the desire to improve, enhance the usefulness and practical ways for mixing paints and painting activities especially on metal, wood, concrete and plastic/acrylic materials, as well as job satisfaction and for economic development.

In this problem, a study that somehow can be used as an alternative for large scale and expensive paint mixer machine available in the market. To make research, do some experiments, share ideas, knowledge, and thoughts in able to develop paint mixer device with painting apparatus as an alternative for large and expensive paint mixer machine: By this information stated, it can help and improve the skills of each individual. This study enhances the performance level of every person who wants to use this product.

A kind of device can be use to provide ease of work in terms of mixing paint using timer with alarm and built-in compressor for spray painting activities. The essence of this research is to add feature of the device and mobility for workmanship, it will enhance construction painter and artists capability to work using paint mixer device with painting apparatus. Perhaps it is difficult as first time to introduce this kind of equipment but to maintain the ability of innovative mind when it comes to technical specification, performance and applicability surely we can able to adapt and fabricate paint mixer device with painting apparatus. Objectives of the Study: Generally, this study aimed to develop paint mixer device with painting apparatus. Specifically, it will sought to; (1) Determine the quality result of develop paint mixer device with painting apparatus in terms of utility, composition and portability. (2) Applicability of develop paint mixer device in terms of design, capacity and efficiency. (3) Acceptability of develop paint mixer device in terms of technical specification and safety features.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Technical specification, Capacity, Efficiency, Design and Paint Mixer

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

The design is actually the product of continuous effort of the researchers to come up with a device suitable for automotive painting services and construction firm activities.

In designing, the researcher will be developing a paint mixer device with painting apparatus with switch timer with alarm, electrical/electronics components and mechanical parts which is properly joint and mechanically fastened with their specific functions. The design is composed of three main parts which utilized the paint mixer, compressor and spray painting accessories which are interconnected to each other whenever the device is used. It also include body frame and caster wheel with lock whenever the device is used. The device can be operational and functional according to each parts and it will be done through manual operation.

The will to create different colours is as old as time. The importance of colour has throughout history been significant on many different levels, mostly it has served as a way for living organism to signal something to its surroundings. For instance, the colour of a plants leaves or petals have been the way with which the plant communicates with its surroundings to attract insects and to spread its DNA, or to signal that it is poisonous. This strategy has also been useful in the same way within the animal kingdom. The colour of an animal can serve as a way to impress the other gender, frighten other animals or be used as camouflage to avoid predators. In other words colours have served a major role in evolution throughout history. This may not have been the case for the human race in terms of biology, although colours have still played a vital role culturally and economically. The colour and design of ones clothes has been, and still to some degree is, a statement and measure of ones' wealth or standing in society. This may not be as apparent in today society but colour serves a big role as a way to express personality and taste. This is not just tied to fashion and clothes, but to all kinds of visual expressions, for example art and furnishing. Paint stores today often use advanced machines to colour match paint. This however, is still a quite involved process requiring several advanced and expensive machines such as spectrophotometers, automatic dispensers and mixers.

Purpose

This report aims to find a colour model that works well when automating the mixing of colours and from that builds a method to do it in practice. It further aims to practically asses the performance of the chosen method. The concept of mixing colours has long seemed like a very practical problem, especially when talking about acrylic paint or other types of paint primarily used in small quantities for applications like art. The trend to miniaturize previously industrial machines for desktop use was an inspiration to see if something similar could be done for automatic colour mixing. Applications, like the ones mentioned earlier, often use other primary.

Mixing colours

There are many different models when it comes to mixing colours. Since the idea of the project is centred on mixing a given colour from an arbitrary set of primary colours and also because the aim at desktop application implies the use of cheap and commonly available sensors and components, three different models were chosen and evaluated throughout the project.

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Types of paint mixing machines available

Generally there are various kinds of paint mixing machines available in the market. They vary in their size, shape, technology and methodologies. Some of the common types of machines are listed below.

Laboratory mixers

These are lab grade machines which are commonly used in laboratories now a days. It depends on high shear lab mixing ideal for research and developmental works. These are used for various kinds of applications such as mixing, emulsifying and dissolving with great precision. Their capacity can vary from 1 ml to 12 liters and offer excellent reproducibility. These are used where process validation is required.

Ultra mix mixers

These are designed for applications which are beyond the capabilities of conventional mixers. They also require a lower shear. These are designed for clean in place and sterilize in place options. The dynamic mixing head provides excellent in tank movement. The large volume of materials is incorporated by a large vortex. This requires low maintenance with robust control process. The design suits excellent chemical services and sanitary requirements.

Inline mixers

These mixers are highly efficient and capable of reducing the mixing time up to a great extent. These can be modified by rapidly interchangeable work heads. This helps to mix, emulsify, homogenize and disperse the colours. The features include aeration free, self-pumping, no bypass and rapid dissolving.

Flash band mixers

These disperse powders into liquids and create a near-perfect consistent homogeneous mixture. This has one of the complex applications. It is a high shear system. It incorporates a wide range of powders. This design helps to incorporate powders on a continuous and semi-continuous basis. This system can also handle a wide range of viscosities. This design is suitable for large production and also is an agglomerate free process.

Bottom entry mixers

These are a series of high shear mixers designed to fit into the bottom of the mixers and sometimes the sides also. These are used coaxially with a slow speed stirrer anchor for high viscous products. The mixer distributes the homogenized output throughout the vessel. This is an ideal option for high viscous products like cosmetics and pharmaceuticals. These can also be used on low viscosity products to wet out powders. It uses a double mechanical shaft for operation.

Dissolver mixers

It uses a powerful and unique mixer present at the bottom of the custom built vessel. The mixer impales a great amount of suction force downwards the liquid surface pulling down the buoyant fluids. These are ripped apart and dissolved throughout the mixture.

Conventional colour mixers have a wide range of industrial applications and is used where there is a requirement of variety of colours with diversified applications. However, with the advancement in technology, more and more industries are leaning towards the use of these machines such as dyeing industries, painting industries etc. Conventional colour mixing and determination process is a gruesome one and also time taking. It requires a lot of effort and consumes a great amount of time. Currently there are large mixers which are used in industries and can only be used in an industrial level. Our aim here is to design and develop such a system which can be used on the individual user level. It has to be a portable one and easy to be operated on. Similarly the function has to be simplified and should be made more user specific. The handling and operation has to be such that a layman would find no difficulty in using it. Very limited work has been done on this level and we aim to take this forward by improving the system performance and the process.

Objectives of the Study:

Generally, this study aimed to develop paint mixer device with painting apparatus. Specifically, it will sought to;

- (1) Determine the quality result of develop paint mixer device with painting apparatus in terms of utility, composition and portability.
- (2) Applicability of develop paint mixer device in terms of design, capacity and efficiency.
- (3) Acceptability of develop paint mixer device in terms of technical specification and safety features.

(11) **Materials and Methods or Methodology** (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The materials used for the development of paint mixer device with painting apparatus were fabricated with the following; 1/4" thick x 24" in width x 42" in length x 32" height x 2" marine plywood supported with 1/2" x 1/2" metal tubular as structural body framing, attached with are two sets of opposite 15" x 22" aluminum sink. The internal component of the device are the following, a 20" x 24" pull and push wooden type drawer use to dip and rub roller brush with paint for painting. Attached also are the circular plastic fiber holder for flexible water hose in order to use for water supply to the lavatory sink. While electrical component of the device for operation are fixed in order for safety of the user such as, a heavy duty breaker, outlet, pdx extension wire connected to the on & off switch

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directed to the #16 flat cord wire for 220volts electrical supply. However, in the operation of the 15 volts portable saw paint mixer motor with cylinder tank 350 max pressure and freon tank 13.6 kg, it is connected to the main on & off switch having the switch timer with alarm also connected to the 18volts transformer, a reverse and forward switch for continuous operation. The built in compressor tank with digital gauge and extra fabricated oxygen tank allowing the air to continuous upon doing spray painting activities.

This study is the developmental method of research, developmental method has been defined as “the systematic study of designing, developing and evaluating instructional programs, processes, and products that must meet the criteria of internal consistency and effectiveness” (Seels & Richey, 1994). The development of paint mixer device with painting apparatus is the modification of the different processes involved in working with spray painting and manual painting activities.

Construction Procedure	Number of months								
	1	2	3	4	5	6	7	8	9
Designing /layout of the device and electrical components and installation	■								
Pre selection of materials		■							
Fabrication			■						
Installation/ Assemble of Components				■					
Checking of structure				■	■				
Construction of the insulation						■			
Product Testing						■			
Testing Revision							■		
Observation Data Recording								■	
Evaluation									■

Location of the study was done in the Automotive and Drafting laboratory room of the Industrial technology department, CapSU Main Campus.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Table 1. Quality result of develop paint mixer device with painting apparatus in terms of utility, composition and portability

Utility	Mean	Verbal Interpretation
1. Planned for a specific purpose in painting and color mixing activities.	4.68	Very Good
2. Useful, especially through being able to perform several functions.	4.37	Very Good
3. Parts and function are commendable for end user (construction painter, mural artists and automotive painting services).	4.45	Very Good
Grand Mean	4.50	Very Good

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Composition		
1. Made of schedule .40mm by 200mm by 200mm metal tubular coated with epoxy primer paint and properly welded parts and joints.	4.41	Very Good
2. Removable 15 volts paint mixer motor integrated with lavatory sink for washing and compartment area for compressor and spray gun.	4.31	Very Good
1. 1/4 thick marine ply board casing with two sliding door and drawer type paint pad for roller paint brush.	4.44	Very Good
Grand Mean	4.39	Very Good
Portability		
1. The device has caster wheel with lock for ease of work.	4.32	Very Good
2. Ability to move or be moved freely and easily.	4.32	Very Good
3. Ability to perform spray painting, roller paint brush and mixing.	4.31	Very Good
Grand Mean	4.31	Very Good

The data showed in **table 1**, Quality result of develop paint mixer device with painting apparatus in terms of **utility**, a grand mean of **4.50** which verbally interpreted as **very good**. This implies that the device have planned for a specific purpose in painting and color mixing activities, useful, especially through being able to perform several functions and parts and function are commendable for end user (construction painter, mural artists and automotive painting services). In terms of **composition**, a grand mean of **4.39** which verbally interpreted as **very good** implies that the device are made of schedule .40mm by 200mm by 200mm metal tubular coated with epoxy primer paint and properly welded parts and joints, removable 15 volts paint mixer motor integrated with lavatory sink for washing and compartment area for compressor and spray gun and 1/4 thick marine ply board casing with two sliding door and drawer type paint pad for roller paint brush..

In terms of **portability**, got a grand mean of **4.31** which verbally interpreted also as **very good** implies that the device having caster wheel with lock for ease of work, ability to move or be moved freely and easily and ability to perform spray painting, roller paint brush and mixing.

Table 2. Applicability of develop paint mixer device in terms of design, capacity and efficiency.

Design	Mean	Verbal Interpretation
1. Layout and plan for fabrication are being followed.	4.40	Very Applicable
2. Fabrication and materials of the device are being specified and measured accurately.	4.35	Very Applicable
3. Packaging are well-covered or decorated with paint.	4.54	Very Applicable
Grand Mean	4.43	Very Applicable
Capacity		
1. A total measurement of 1.50 meters in width, 2.00 meters in length and 2.80 meters in height with a load of 50 kilos.	4.40	Very Applicable
2. A paint mixer device can operate 3 to 5 minutes with the capacity of 2 to 3 liters of paint for mixing color.	4.35	Very Applicable
3. Approximate 2 person can use or utilize the device with 4 to 8 hours a day operation.	4.45	Very Applicable
Grand Mean	4.40	Very Applicable
Efficiency		

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1. Minimum of 5 to 10 minutes to assemble with a maximum of 1 person skilled/not skilled in the art with proper orientation.	4.40	Very Applicable
2. Capable to mixed 5 to 12 liters or 3 gallons of colors in 1 hour a day with the span of 3 meters height to 4 meters wide concrete wall.	4.35	Very Applicable
3. Effective 200 to 250 psi (pound per square inch) with 125ml epoxy primer paint inside the spray gun aluminum bottle cup in the maximum of 10 minutes while 3 to 4 hours using roller type brush in a span 4 meters Concrete wall.	4.38	Very Applicable
Grand Mean	4.38	Very Applicable

The data showed in **table 2**, Applicability of develop paint mixer device with painting apparatus in terms of **design**, a grand mean of **4.43** which verbally interpreted as **very applicable**. This implies that the device layout and plan for fabrication are being followed, fabrication and materials of the device are being specified and measured accurately and packaging are well-covered or decorated with paint. In terms of **capacity**, a grand mean of **4.40** which verbally interpreted as **very applicable** implies that a total measurement of 1.50 meters in width, 2.00 meters in length and 2.80 meters in height with a load of 50 kilos also paint mixer device can operate 3 to 5 minutes with the capacity of 2 to 3 liters of paint for mixing color with approximate 2 person can use or utilize the device with 4 to 8 hours a day operation. In terms of **efficiency**, got a grand mean of **4.38** which verbally interpreted also as **very applicable** implies that the device have a minimum of 5 to 10 minutes to assemble with a maximum of 1 person skilled/not skilled in the art with proper orientation, capable to mixed 5 to 12 liters or 3 gallons of colors in 1 hour a day with the span of 3 meters height to 4 meters wide concrete wall and effective 250 psi (pound per square inch) with 125ml epoxy primer paint inside the spray gun aluminum bottle cup in the maximum of 10 minutes while 3 to 4 hours using roller type brush in a span 4 meters concrete wall.

Table 3. Acceptability of paint mixer device with painting apparatus in terms of technical specification and feature of the device.

Technical specification and feature of the device	Mean	Verbal Interpretation
1.15 volts adjustable, removable paint mixer motor integrated with lavatory sink for washing and compartment area for compressor and spray gun coated with acrylic lacquer paint for water resistant.	4.53	Very Acceptable
2. Fabricated with .04mm thickness by .50 cm by .50 cm metal tubular body framing and 1/4 thick marine ply board casing with two sliding door and drawer type paint pad for roller paint brush, attached also are 2 1/2" caster wheel for ease of work.	4.30	Very Acceptable
3) Installed with 20 watts breaker, convenience outlet for compressor connected to switch on/off with 18 watts transformer and exhaust fan and #16 flat cord extension wire to be connected to 220 volts as main supply.	4.60	Very Acceptable
4) Fabricated with 12" rounded, rotatable fiber acrylic plastic for water flexible hose connected to		

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lavatory sink and faucet for washing and cleaning of brushes and discarded paint.	4.48	Very Acceptable
5)Attached are 13" in length and 18" in wide adjustable drawer type and easy to use roller paint pad integrated with modified compressor motor with Freon tank and spray gun for painting activities.	4.35	Very Acceptable

Grand Mean	4.45	Very Acceptable
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The data showed in **table 3**, that the the acceptability of develop paint mixer device with painting apparatus considering its technical specification and feature of the device got a grand mean of **4.45** which verbally interpreted as **very acceptable**. This implies that fabricated with .04mm thickness by .50 cm by .50 cm metal tubular body framing and 1/4 thick marine ply board casing with two sliding door and drawer type paint pad for roller paint brush, attached also are 2 1/2" caster wheel for ease of work got the lowest mean of **4.30** which verbally interpreted also as **very acceptable**. While, Installed with 20 watts breaker, convenience outlet for compressor connected to switch on/off with 18 watts transformer and exhaust fan and #16 flat cord extension wire to be connected to 220 volts as main supply got a highest mean of **4.60** verbally interpreted as **very acceptable**.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Based on the findings of the study, the following conclusions were drawn:

The quality result of develop paint mixer device with painting apparatus in terms of utility, composition and portability have planned for a specific purpose in painting and color mixing activities and properly welded parts, joints, removable paint mixer motor integrated with lavatory sink, compressor and spray component with drawer type paint pad for roller paint brush and ability to perform spray painting, roller paint brush and mixing.

Moreover, the applicability of the device in terms of design, capacity and efficiency derived to a well-planned layout for fabrication, can operate both mixing of colors, spray painting process and manual utilization of roller type brush for wall painting with effective hands on activity for a period of time which will benefit the consumer or end user.

While acceptability of paint mixer device with painting apparatus are very acceptable for the whole process of painting process through manual or spray painting with proper orientation of its technical specification and feature of the said device.

Recommendations

Based on the findings and conclusions of the study, the following recommendations are forwarded.

Development of paint mixer device with painting apparatus is recommended to enhance color mixing and spray painting processes using 18 volts motor and to utilize some of its body parts for faster and more effective performance while doing painting and mixing process. Therefore, it may suggest that fabrication of the device will be more durable if material used will be aluminum tubular type framing for table, plain galvanized iron sheet for packaging or stainless sheet. Further innovation, includes using automatic computer system attach to the table covered with pull-up type cover for safety in operation.

Further study is also recommended that other material except wooden framing, fibre optics or aluminium cladding and moulding should be utilized to create this tool. Further study is encouraged also to improve the features of the device that could be added or tested to enhance the design of the product.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

The device is being develop to come up with usefulness, portability and convenient while at work. Some of its function are integrated in order to enhance workmanship and accuracy which is limited only for manual painting and spray painting.

The development of this device is to produced very good and effective result in terms of wall/mural painting and furniture or metal body painting.

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A device wherein color mixing process and washing/cleaning of painting apparatus is involved for economical application in terms of roller painting application which also add aesthetic function to enhance wall painting and spray painting.

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

This product relates to a paint mixer device with painting apparatus particularly to color mixing and wall painting. This device is an alternative to conventional single unit of paint mixer device or machine which often used in subject area of visual techniques and architectural design, automotive and car servicing.

It furthers focused on the mechanism of its parts and function, acceptability of the said device enhance the performance level of the user for project making.

The processes or services transferred as extension activities will be part of the Industrial technology program and adapt a barangay program of the institution. The project is intended to equip drafting student, visual artist, painter for livelihood skills for those out of school youth, by stander, construction worker, laborer and they in turn will become a part of community livelihood program.

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

Unit	Quantity	Description	Unit price	Amount
1	Sheet	1/4" x 4' x 8' Marine Plyboard	450	450
1	Sheet	3/4 x 4 x 8 Marine Plyboard	850	850
2	Pieces	1/2 x 1/2 x 3meters metal tubular	600	1200
1	set	1/2 x 1/2 x 3meters Angle bar	700	700
2	Set	15 x 20 Aluminum Lavatory Sink	500	1000
1	Unit	18 volts Motor	800	800
1	Unit	On/Off switch	300	300
1	Unit	Breaker	400	400
6	Meters	#16 flat cord	150	150
4	Set	2" Caster wheel	75	300
1/2 kilo	Pieces	Blind rivets 1/8 x 1 1/2	50	50
2	liters	Epoxy primer	150	300
10	Pieces	Bolt and Nut	25	250
2	Set	Eagle type outlet	150	300
		Total Amount		7,050

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		7,050 (0.40) =Labor (2,820.00)		
		Summary of Expenses		9,870

End user of the technology are those in the field of Engineering and Construction Firm, Junior-Senior High vocational technology-Drafting Major/Related Industrial Arts subject, and Department of Public Works and Highway Engineering Department.

(17) **Acknowledgments** (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) **References** (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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Design and Manufacturing of Paint Mixture Equipment in Core Manufacturing Industry.
Shital Bhosale,Nawghane Rohan, Patil Akshaykumar,Patil Sujit,Thorave Swapnil,Mechanical Engineering Dept, JSPM Narhe Technical Campus, Pune, India© 2021 JETIR July 2021, Volume 8, Issue 7 www.jetir.org (ISSN-2349-5162)

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